1 I

1.1 (1)

From the equation of motion,

$$m_i \frac{d\vec{v}_i}{dt} = q_i \left(\vec{E}(\vec{x}_i, t) + \vec{v}_i \times \vec{B}(\vec{x}_i, t) \right),$$

we obtain

$$\begin{aligned} \frac{dK_i}{dt} &= \frac{d}{dt}\left(\frac{1}{2} m_i |\vec{v}_i|^2 \right) = m_i \vec{v}_i \cdot \frac{d\vec{v}_i}{dt} \\ &= q_i \vec{v}_i \cdot \left(\vec{E}(\vec{x}_i, t) + \vec{v}_i \times \vec{B}(\vec{x}_i, t) \right) \\ &= q_i \vec{v}_i \cdot \vec{E}(\vec{x}_i, t), \end{aligned}$$

since $\vec{v}_i \cdot (\vec{v}_i \times \vec{B}) = 0$.

1.2 (2)

Using $W = \sum_i K_i$ and

$$\vec{J}(\vec{x}, t) = \sum_i q_i \vec{v}_i \delta(\vec{x} - \vec{x}_i),$$

we have

$$\begin{aligned} \frac{dW}{dt} &= \sum_i \frac{dK_i}{dt} = \sum_i q_i \vec{v}_i \cdot \vec{E}(\vec{x}_i, t) \\ &= \sum_i \int_V \vec{E}(\vec{x}, t) \cdot q_i \vec{v}_i \delta(\vec{x} - \vec{x}_i)\, dV \\ &= \int_V \vec{E}(\vec{x}, t) \cdot \vec{J}(\vec{x}, t)\, dV. \end{aligned}$$

1.3 (3)

From Maxwell's equations,

$$\nabla \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}, \qquad \nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}.$$

Using the vector identity

$$\nabla \cdot (\vec{E} \times \vec{H}) = \vec{H} \cdot (\nabla \times \vec{E}) - \vec{E} \cdot (\nabla \times \vec{H}),$$

we find

$$\begin{aligned} \vec{E} \cdot \vec{J} &= \vec{E} \cdot (\nabla \times \vec{H}) - \vec{E} \cdot \frac{\partial \vec{D}}{\partial t} \\ &= \vec{H} \cdot (\nabla \times \vec{E}) - \nabla \cdot (\vec{E} \times \vec{H}) - \vec{E} \cdot \frac{\partial \vec{D}}{\partial t} \\ &= -\vec{H} \cdot \frac{\partial \vec{B}}{\partial t} - \vec{E} \cdot \frac{\partial \vec{D}}{\partial t} - \nabla \cdot (\vec{E} \times \vec{H}) \\ &= -\frac{\partial}{\partial t}\left[\frac{1}{2}(\vec{E} \cdot \vec{D} + \vec{H} \cdot \vec{B}) \right] - \nabla \cdot (\vec{E} \times \vec{H}), \end{aligned}$$

where we have used $\vec{D} = \epsilon_0 \vec{E}$ and $\vec{B} = \mu_0 \vec{H}$, with constant ϵ_0 and μ_0.

Thus, the electromagnetic energy density and Poynting vector are

$$u = \frac{1}{2}(\vec{E}\cdot\vec{D} + \vec{H}\cdot\vec{B}), \qquad \vec{S} = \vec{E}\times\vec{H}.$$

Integrating over the volume and using (2), we obtain

$$\frac{dW}{dt} + \frac{d}{dt}\int_V u\,dV + \int_V \nabla\cdot\vec{S}\,dV = 0.$$

1.4 (4)

The first term, dW/dt, represents the rate of change of the kinetic energy of the charged particles.

The second term, $d(\int_V u\,dV)/dt$, represents the rate of change of the energy stored in the electromagnetic field.

The third term, $\int_V \nabla\cdot\vec{S}\,dV$, represents the net outward flow of electromagnetic energy through the boundary of the chosen volume.

2 II

2.1 (5)

For the plane wave

$$\vec{E} = E_0\sin(\omega t - kz)\vec{e}_x,$$

Faraday's law gives

$$\nabla\times\vec{E} = -kE_0\cos(\omega t - kz)\vec{e}_y = -\frac{\partial\vec{B}}{\partial t}.$$

Hence,

$$\vec{B} = \frac{k}{\omega}E_0\sin(\omega t - kz)\vec{e}_y = \frac{1}{c}\vec{e}_z\times\vec{E},$$

and

$$\vec{H} = \frac{\vec{B}}{\mu_0} = \sqrt{\frac{\epsilon_0}{\mu_0}}E_0\sin(\omega t - kz)\vec{e}_y.$$

The Maxwell–Ampère equation also yields the dispersion relation

$$\omega^2 = \frac{k^2}{\epsilon_0\mu_0}, \qquad \frac{\omega}{k} = c = \frac{1}{\sqrt{\epsilon_0\mu_0}}.$$

2.2 (6)

The Poynting vector is

$$\vec{S} = \vec{E}\times\vec{H} = \sqrt{\frac{\epsilon_0}{\mu_0}}E_0^2\sin^2(\omega t - kz)\vec{e}_z.$$

The electromagnetic energy density is

$$\begin{aligned} u &= \frac{1}{2}(\vec{E}\cdot\vec{D} + \vec{H}\cdot\vec{B}) \\ &= \epsilon_0 E_0^2\sin^2(\omega t - kz). \end{aligned}$$

Since the time average of $\sin^2(\omega t - kz)$ over one period is $1/2$,

$$\langle S\rangle = \frac{1}{2}\sqrt{\frac{\epsilon_0}{\mu_0}}E_0^2, \qquad \langle u\rangle = \frac{1}{2}\epsilon_0 E_0^2.$$

Therefore,

$$\langle S\rangle = \alpha\langle u\rangle, \qquad \alpha = \frac{1}{\sqrt{\epsilon_0\mu_0}} = c.$$

3 III

3.1 (7)

Taking $\Delta\phi = \phi(A_1) - \phi(A_2) > 0$, the electric field on the wire's surface is

$$\vec{E} = \frac{\Delta\phi}{L}\vec{e}_z.$$

From Ampère's law around the circular cross section,

$$\oint \vec{H}\cdot d\vec{\ell} = H_\theta(2\pi a) = J\pi a^2.$$

Thus,

$$\vec{H} = \frac{Ja}{2}\vec{e}_\theta.$$

The Poynting vector on the surface is

$$\begin{aligned}\vec{S} &= \vec{E}\times\vec{H}\\ &= \frac{\Delta\phi}{L}\frac{Ja}{2}(\vec{e}_z\times\vec{e}_\theta)\\ &= -\frac{\Delta\phi Ja}{2L}\vec{e}_r.\end{aligned}$$

3.2 (8)

Using the divergence theorem,

$$\begin{aligned}\int_V \nabla\cdot\vec{S}\,dV &= \oint_{\partial V}\vec{S}\cdot d\vec{A}\\ &= \int_{\text{lateral surface}} S_r\,dA\\ &= \left(-\frac{\Delta\phi Ja}{2L}\right)(2\pi aL)\\ &= -\pi a^2 J\Delta\phi.\end{aligned}$$

The flux through the two end faces vanishes because $\vec{S}$ is radial. The negative sign means that electromagnetic energy flows into the wire through its lateral surface.

For steady electromagnetic fields, the field-energy term vanishes. The energy-conservation equation then gives

$$\frac{dW}{dt} = \pi a^2 J\Delta\phi = I\Delta\phi, \qquad I = J\pi a^2.$$

Thus, the inward electromagnetic energy flux is balanced by the work done on the charged particles. In a real resistive conductor, this power is transferred to the material as Joule heat.